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FRONT STRUCTURE OF VEHICLE

Abstract

A front structure of a vehicle has a pair of vehicle body frames, a pair of crash boxes, a support member, and a bumper reinforcement. The vehicle body frames are disposed at an interval in a width direction of the vehicle. The crash boxes are located in front of the vehicle body frames to be attached to the vehicle body frames. The support member includes a support body located between the crash boxes, and a pair of extension portions extending outward in the width direction from the support body on both sides. The bumper reinforcement extends in the width direction to be attached to front ends of the crash boxes. The extension portion includes a guide portion which guides the crash box to an attachment position by engaging with the crash box. The guide portion has a surface inclined with respect to a front-rear direction of the vehicle.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2024-025341 filed on Feb. 22, 2024, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a front structure of a vehicle.

BACKGROUND ART

[0003] JP2019-059415A describes a front structure of a vehicle. The front structure of a vehicle described in JP2019-059415A includes a pair of front side members located on both sides in a vehicle width direction, and a pair of crash boxes coupled to respective front ends of the pair of front side members. The front structure of a vehicle further includes a bumper reinforcement coupled to front ends of the pair of crash boxes. The crash boxes each include a tubular main body portion, a first coupling piece that protrudes forward from an upper edge of a front end portion of the main body portion, and a second coupling piece that protrudes downward from a lower edge of the front end portion of the main body portion. A second flange portion is provided at a rear end portion of the main body portion. A second upper hole portion and a second lower hole portion are respectively formed on the first coupling piece and the second coupling piece. First upper holes and first lower holes, which respectively communicate with the second upper holes and the second lower holes, are formed on the bumper reinforcement. Nuts are respectively screwed to bolts inserted through the first upper holes and the second upper holes and bolts inserted through the first lower holes and the second lower holes, whereby the bumper reinforcement is fastened to the pair of crash boxes. The second flange portion is fastened by the bolts to a first flange portion provided at a front end portion of the front side member.

[0004] In the front structure of the vehicle in the related art including the structure described in JP2019-059415A, variations in processing accuracy and assembly accuracy of the front side members and variations in processing accuracy and assembly accuracy of the crash boxes occur. Therefore, the first upper holes and the first lower holes of the bumper reinforcement may be displaced with respect to the second upper holes and the second lower holes of the crash boxes, which may make assembly difficult.

[0005] Therefore, the bumper reinforcement is assembled to the crash boxes by setting the first upper holes and the first lower holes larger than the second upper holes and the second lower holes in consideration of the above-described variations. However, even in this case, variations in positions of the front ends of the crash boxes with respect to the bumper reinforcement are still large.

[0006] Such a circumstance is more remarkable in a vehicle such as a battery electric vehicle (BEV) in which lengths in a front-rear direction of the crash boxes are large.

SUMMARY OF INVENTION

[0007] An aspect of the present disclosure will be described.

[0008] (1) The aspect of the present disclosure provides a front structure of a vehicle, including:

[0009] a pair of vehicle body frames disposed at an interval in a vehicle width direction; [0010] a pair of crash boxes located in front of the vehicle body frames in a front-rear direction of the vehicle, and attached to the vehicle body frames respectively; [0011] a support member including a support body located between the pair of crash boxes, and a pair of extension portions that extend outward in the vehicle width direction from the support body and are located on both sides of the support body in the vehicle width direction; and [0012] a bumper reinforcement that extends in the vehicle width direction, and is attached to front ends of the pair of crash boxes, [0013] in which the

extension portion includes a guide portion configured to guide the crash box to an attachment position by engaging with a front end of the crash box, the guide portion having an inclined surface that is inclined with respect to the front-rear direction.

[0014] According to the above-described configuration, the guide portion is engaged with the front end of the crash box, so that a position of the crash box is guided to an attachment position along the inclined surface of the guide portion. Accordingly, a variation in a position of the front end of the crash box, that is, a position of a portion of the crash box to which the bumper reinforcement is attached, is prevented. Therefore, the crash boxes are easily assembled to predetermined positions of the bumper reinforcement.

[0015] (2) In the front structure of the aspect, the crash box includes a tubular portion extending in the front-rear direction, and a coupling portion coupled to a front end of the tubular portion and protruding to an outer peripheral side of the tubular portion, [0016] a front end surface of the coupling portion has two engaged portions having a hole shape or an aperture shape and provided at an interval in the vehicle width direction, and [0017] two of the guide portions are inserted into the two engaged portions respectively.

[0018] According to the above-described configuration, the two guide portions are inserted into the two engaged portions having the hole shape or the aperture shape, so that the crash box is guided to the attachment position along the inclined surface in the vehicle width direction. Further, the two guide portions are engaged with the two engaged portions, so that a relative movement of the crash box with respect to the support member is restricted. Therefore, the crash box can be fixed at the attachment position.

[0019] (3) In the front structure of the aspect, the crash box includes a tubular portion extending in the front-rear direction, and a coupling portion coupled to a front end of the tubular portion, [0020] the coupling portion has a cutout portion not protruding outward in a radial direction of the tubular portion, and [0021] the guide portion has a pair of the inclined surfaces configured to sandwich the cutout portion from an inner peripheral side and an outer peripheral side of the tubular portion in the vehicle width direction, and a facing surface facing a front end of the cutout portion.

[0022] According to the above-described configuration, the guide portion is engaged with the cutout portion of the coupling portion, so that the crash box is guided to an attachment position along the inclined surface in the vehicle width direction. Further, the cutout portion is surrounded by the pair of inclined surfaces and the facing surface, so that a relative movement of the crash box with respect to the support member is restricted. Therefore, the crash box can be fixed at the attachment position.

[0023] (4) In the front structure of the aspect, the support member including the support body and the pair of extension portions is integrally molded.

[0024] According to the above-described configuration, since the members constituting the support member are integrally molded, the accuracy of a shape of the support member is improved. Therefore, the guide to the attachment position implemented by the guide portions is accurately performed.

[0025] (5) In the front structure of the aspect, the support member is made of a fiber reinforced resin.

[0026] According to the above-described configuration, by using the support member made of the fiber reinforced resin, it is possible to reduce a weight of the support member while maintaining a strength of the support member.

[0027] (6) In the front structure of the aspect, the crash box includes a tubular portion extending in the front-rear direction, and a coupling portion coupled to a front end of the tubular portion and protruding to an outer peripheral side of the tubular portion, and [0028] a front end surface of the coupling portion has an accommodating recessed portion configured to accommodate the extension portion.

[0029] According to the above-described configuration, in the case where the extension portion

made of the resin is located between the coupling portion, which is the front end of the crash box, and the bumper reinforcement, the extension portion is accommodated in the accommodating recessed portion. Accordingly, a compressive stress acting on the extension portion sandwiched between the crash box and the bumper reinforcement can be reduced.

[0030] Therefore, creep deformation of the extension portion is prevented. Therefore, the durability of the support member can be improved.

[0031] (7) In the front structure of the aspect, a rear surface of the bumper reinforcement has an accommodating recessed portion configured to accommodate the extension portion.

[0032] According to the above-described configuration, in the case where the extension portion made of the resin is located between the coupling portion, which is the front end of the crash box, and the bumper reinforcement, the extension portions are accommodated in the accommodating recessed portion. Accordingly, a compressive stress acting on the extension portion sandwiched between the crash box and the bumper reinforcement can be reduced. Therefore, creep deformation of the extension portion is prevented. Therefore, the durability of the support member can be improved.

[0033] According to the aspect of the present disclosure, the crash boxes are easily assembled to predetermined positions of the bumper reinforcement.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0034] Exemplary embodiment(s) of the present invention will be described in detail based on the following figures, wherein:

[0035] FIG. 1 is an exploded perspective view illustrating a front structure of a vehicle according to a first embodiment;

[0036] FIG. 2 is a cross-sectional view taken along a line 2-2 in FIG. 1;

[0037] FIG. 3 is a cross-sectional view illustrating a state in which a crash box is engaged with an inclined surface of a second guide portion;

[0038] FIG. 4 is an exploded perspective view illustrating a front structure of a vehicle according to a second embodiment;

[0039] FIG. 5 is a cross-sectional view taken along a line 5-5 in FIG. 4;

[0040] FIG. 6 is a cross-sectional view illustrating a state in which a crash box is engaged with inclined surfaces of a guide portion;

[0041] FIG. 7 is a perspective view illustrating a modification of a support member; and

[0042] FIG. 8 is a cross-sectional view taken along a line 8-8 in FIG. 7.

DESCRIPTION OF EMBODIMENTS

First Embodiment

[0043] Hereinafter, the first embodiment of a front structure of a vehicle will be described with reference to FIGS. 1 to 3.

[0044] As illustrated in FIG. 1, the front structure of a vehicle includes a pair of front side members 10, a pair of crash boxes 20, a support member 30, and a bumper reinforcement 50.

[0045] In the following description, a front-rear direction of the vehicle is referred to as a front-rear direction L, a vehicle width direction is referred to as a vehicle width direction W, and an upper-lower direction of the vehicle when the vehicle is located on a horizontal plane is referred to as an upper-lower direction Z.

[0046] Further, a front side and a rear side in the front-rear direction L are simply referred to as a “front side” and a “rear side” respectively.

[0047] The vehicle according to the present embodiment has a symmetrical structure in the vehicle width direction W. Therefore, hereinafter, the left half of the vehicle at the time when viewed from

the front to the rear will be described, and a description for the right half may be omitted.

<Front Side Member **10**>

[0048] As illustrated in FIG. **1**, the pair of front side members **10** are tubular and extend in the front-rear direction L, and are disposed at an interval in the vehicle width direction W.

[0049] A vehicle body-side flange portion **11** having a rectangular annular shape in a front view, which protrudes outward in a radial direction of the front side member **10**, is provided at a front end of the front side member **10**. A through hole **11a**, which penetrates the vehicle body-side flange portion **11** in the front-rear direction L, is formed at each of four corner portions of the vehicle body-side flange portion **11**.

<Crash Box **20**>

[0050] As illustrated in FIG. **1**, the pair of crash boxes **20** are located in front of the front side members **10** and are attached to the front side members **10** respectively.

[0051] The crash box **20** includes a tubular portion **21** extending in the front-rear direction L, a tubular portion-side flange portion **22** coupled to a rear end of the tubular portion **21**, and a coupling portion **23** coupled to a front end of the tubular portion **21**.

[0052] The tubular portion-side flange portion **22** protrudes outward in a radial direction of the tubular portion **21** from the rear end of the tubular portion **21** and has a rectangular annular shape in a rear view.

[0053] On the tubular portion-side flange portion **22**, through holes **22a** are formed at positions corresponding to the through holes **11a** of the vehicle body-side flange portion **11**.

[0054] The crash box **20** is attached to the front end of the front side member **10** by screwing nuts **16** to respective bolts **15** inserted through the through holes **22a** and **11a**.

[0055] The coupling portion **23** protrudes to an outer peripheral side of the tubular portion **21** and has a rectangular plate shape in a front view. The coupling portion **23** has a pair of long sides **24a** and a pair of short sides **24b**.

[0056] The front end of the tubular portion **21** is welded to a rear end surface of the coupling portion **23**.

[0057] The coupling portion **23** has a front end surface **24** that is a surface opposite to the rear end surface in the front-rear direction L.

[0058] As illustrated in FIGS. **1** and **2**, the front end surface **24** includes a flat portion **25** and an accommodating recessed portion **26** that is recessed rearward with respect to the flat portion **25**.

[0059] As illustrated in FIG. **1**, the flat portion **25** is formed along the pair of long sides **24a** and the short side **24b** on an outer side in the vehicle width direction W.

[0060] The accommodating recessed portion **26** is formed by cutting out the front end surface **24**, and opens inward in the vehicle width direction W and forward.

[0061] A bottom surface **26b** of the accommodating recessed portion **26** has a rectangular shape in a front view.

[0062] A through hole **26a**, which penetrates the accommodating recessed portion **26** in the front-rear direction L, is formed at each of four corner portions of the bottom surface **26b** of the accommodating recessed portion **26**.

[0063] Two holes **27** are formed at an interval in the vehicle width direction W on a bottom surface **26b** of the accommodating recessed portion **26**.

[0064] As illustrated in FIG. **2**, the two holes **27** each have a rectangular shape in a front view, extend in the upper-lower direction Z, and have an inner peripheral edge **27a** and an outer peripheral edge **27b** that are formed at an interval in the vehicle width direction W.

[0065] The inner peripheral edge **27a** is located more inward of the tubular portion **21** in the radial direction than an outer peripheral surface of the tubular portion **21**.

[0066] The outer peripheral edge **27b** is located more outward of the tubular portion **21** in the radial direction than the outer peripheral surface of the tubular portion **21**.

[0067] The crash box **20** is made of a metal.

<Support Member 30>

[0068] As illustrated in FIGS. 1 and 2, the support member 30 is attached to the pair of crash boxes 20 to maintain a posture of the pair of crash boxes 20.

[0069] The support member 30 is made of a fiber reinforced resin. A reinforcing fiber contained in the fiber reinforced resin is, for example, a glass fiber.

[0070] As illustrated in FIG. 1, the support member 30 includes a support body 31 located between the pair of crash boxes 20 and a pair of extension portions 35 that extend outward in the vehicle width direction W from the support body 31 and are located on both sides of the support body 31 in the vehicle width direction W.

[0071] In the present embodiment, the support body 31 and the pair of extension portions 35 are integrally molded by injection molding.

(Support Body 31)

[0072] As illustrated in FIG. 1, the support body 31 includes an upper wall portion 32 extending in the vehicle width direction W, a pair of side wall portions 33 extending downward from both ends in the vehicle width direction W of the upper wall portion 32, and a pair of inclined portions 34 extending from lower ends of the pair of side wall portions 33 toward a central portion of the upper wall portion 32.

[0073] A headlight and a hood lock for a bonnet hood (both not illustrated) are attached to the upper wall portion 32.

[0074] The inclined portion 34 is inclined such that the more inward a portion is located in the vehicle width direction W, the higher the portion is.

(Extension Portion 35)

[0075] As illustrated in FIG. 1, the extension portion 35 includes a first portion 36 extending forward from a lower end of the side wall portion 33 and a second portion 37 extending outward in the vehicle width direction W from a front end of the first portion 36.

[0076] The first portion 36 is located more inward than the crash box 20 in the vehicle width direction W. The front end of the first portion 36 is disposed side by side with the coupling portion 23 in the vehicle width direction W.

[0077] As illustrated in FIG. 2, the second portion 37 includes a plate-shaped base portion 38 extending outward in the vehicle width direction W from the front end of the first portion 36, and a first guide portion 40 and a second guide portion 45 that protrude rearward from the base portion 38.

[0078] The base portion 38 is accommodated in the accommodating recessed portion 26 of the coupling portion 23.

[0079] An outer edge of the base portion 38 extends along an inner edge of the accommodating recessed portion 26 of the coupling portion 23 (see FIG. 1).

[0080] A front surface of the base portion 38 is located on the same plane as the flat portion 25 of the front end surface 24.

[0081] On the base portion 38, a through hole 38a is formed at each of positions corresponding to the through holes 26a.

(First Guide Portion 40 and Second Guide Portion 45)

[0082] As illustrated in FIG. 2, the first guide portion 40 and the second guide portion 45 are provided at positions corresponding to the holes 27.

[0083] The first guide portion 40 and the second guide portion 45 are arranged in order from the outer side in the vehicle width direction W.

[0084] The first guide portion 40 includes a first wall portion 41 and a second wall portion 42 that face each other in the vehicle width direction W.

[0085] The first wall portion 41 protrudes rearward from the base portion 38 along the front-rear direction L. A rear surface of the first wall portion 41 is in contact with an inner peripheral surface of the hole 27.

[0086] The second wall portion **42** is located more inward than the first wall portion **41** in the vehicle width direction **W** and protrudes from the base portion **38** toward a rear end of the first wall portion **41**. The second wall portion **42** is inclined such that the more rearward a portion is located, the more outward the portion is located in the vehicle width direction **W**. A rear surface of the second wall portion **42** is an inclined surface **42a** which is inclined with respect to both the front-rear direction **L** and the vehicle width direction **W**, and is in contact with the front end of the tubular portion **21**.

[0087] The second guide portion **45** includes a third wall portion **46** and a fourth wall portion **47** that face each other in the vehicle width direction **W**.

[0088] The third wall portion **46** is inclined such that the more rearward a portion is located, the more outward the portion is located in the vehicle width direction **W**. A rear surface of the third wall portion **46** is an inclined surface **46a** which is inclined with respect to both the front-rear direction **L** and the vehicle width direction **W**.

[0089] The fourth wall portion **47** is located more outward than the third wall portion **46** in the vehicle width direction **W** and protrudes from the base portion **38** toward a rear end of the third wall portion **46**. The fourth wall portion **47** is inclined such that the more rearward a portion is located, the more inward the portion is located in the vehicle width direction **W**. A rear surface of the fourth wall portion **47** is an inclined surface **47a** which is inclined with respect to both the front-rear direction **L** and the vehicle width direction **W**, and is in contact with the front end of the tubular portion **21**.

<Bumper Reinforcement **50**>

[0090] As illustrated in FIGS. **1** and **2**, the bumper reinforcement **50** has a plate shape, extends in the vehicle width direction **W**, and is located in front of the pair of crash boxes **20**.

[0091] On the bumper reinforcement **50**, a through hole **50a** is formed at each of positions corresponding to the through holes **38a** of the base portion **38**.

[0092] The bumper reinforcement **50** is attached to front ends of the pair of crash boxes **20** by screwing nuts **18** to respective bolts **17** inserted through the through holes **50a**, **38a**, and **26a**.

[0093] The bumper reinforcement **50** is made of a metal.

[0094] In the present embodiment, the pair of front side members **10** correspond to a pair of vehicle body frames according to the present invention. In the present embodiment, the hole **27** corresponds to an engaged portion according to the present invention.

Functions of First Embodiment

[0095] As illustrated in FIG. **3**, the second guide portion **45** is engaged with the front end of the crash box **20**, so that a position of the crash box **20** is guided to an attachment position along the inclined surface **46a** of the second guide portion **45**. Accordingly, a variation in a position of the front end of the crash box **20**, that is, a position of a portion of the crash box **20** to which the bumper reinforcement **50** is attached, is prevented.

Effects of First Embodiment

[0096] (1-1) The extension portion **35** includes the first guide portion **40** and the second guide portion **45** configured to guide the crash box **20** to the attachment position by engaging with the front end of the crash box **20**. The first guide portion **40** and the second guide portion **45** respectively have the inclined surfaces **42a** and **46a** which are inclined with respect to both the front-rear direction **L** and the vehicle width direction **W**.

[0097] According to the configuration, the above-described function is implemented, so that the crash boxes **20** are easily assembled to predetermined positions of the bumper reinforcement **50**. Accordingly, the bumper reinforcement **50** can be reliably attached to the crash boxes **20**.

[0098] (1-2) The front end surface **24** of the coupling portion **23** has two holes **27** provided at an interval in the vehicle width direction **W**. The first guide portion **40** and the second guide portion **45** are respectively inserted into the two holes **27**.

[0099] According to the configuration, the first guide portion **40** and the second guide portion **45**

are respectively inserted into the two holes **27**, so that the crash box **20** is guided to the attachment position along the inclined surfaces **42a** and **46a** in the vehicle width direction W. Further, the first guide portion **40** and the second guide portion **45** are respectively engaged with the two holes **27**, so that a relative movement of the crash box **20** with respect to the support member **30** is restricted. Therefore, the crash box **20** can be fixed at the attachment position.

[0100] (1-3) The support member **30** is integrally molded.

[0101] According to such a configuration, since members constituting the support member **30** are integrally molded, the accuracy of a shape of the support member **30** is improved. Therefore, the guide to the attachment position implemented by the first guide portion **40** and the second guide portion **45** is accurately performed.

[0102] (1-4) The support member **30** is made of the fiber reinforced resin.

[0103] According to the configuration, by using the support member **30** made of the fiber reinforced resin, it is possible to reduce a weight of the support member **30** while maintaining a strength of the support member **30**.

[0104] (1-5) The front end surface **24** of the coupling portion **23** has the accommodating recessed portion **26** configured to accommodate the extension portion **35**.

[0105] According to the configuration, in the case where the extension portion **35** made of the resin is located between the coupling portion **23**, which is the front end of the crash box **20**, and the bumper reinforcement **50**, the extension portion **35** is accommodated in the accommodating recessed portion **26**. Accordingly, a compressive stress acting on the extension portion **35** sandwiched between the crash box **20** and the bumper reinforcement **50** can be reduced. Therefore, creep deformation of the extension portion **35** is prevented. Therefore, the durability of the support member **30** can be improved.

Second Embodiment

[0106] Next, the second embodiment of the front structure of a vehicle will be described with reference to FIGS. **4** to **6**, focusing on differences from the first embodiment.

[0107] In the following description, configurations in the second embodiment that are the same as or correspond to those in the first embodiment are denoted by reference numerals obtained by adding “50” to reference numerals “*” of the configurations in the first embodiment, and redundant description may be omitted.

<Crash Box **70**>

[0108] As illustrated in FIG. **4**, a coupling portion **73** having a rectangular annular shape in a front view is coupled to a front end of a tubular portion **71**. The coupling portion **73** has a protruding portion **73a** protruding outward in a radial direction of the tubular portion **71** and a cutout portion **73b** not protruding outward in the radial direction of the tubular portion **71**.

[0109] In the present embodiment, the front end of the tubular portion **71** has a square shape in a front view. The front end of the tubular portion **71** extends in the vehicle width direction W, and has an upper edge **71a** and a lower edge **71b** formed at an interval in the upper-lower direction Z, and a pair of side edges **71c** that connects end portions of the upper edge **71a** and respective end portions of the lower edge **71b** in the vehicle width direction W.

[0110] The protruding portion **73a** is coupled to the upper edge **71a**, the lower edge **71b**, and

[0111] the side edge **71c** on an outer side in the vehicle width direction W of the pair of side edges **71c** of the front end of the tubular portion **71**.

[0112] The cutout portion **73b** is coupled to the side edge **71c** on an inner side in the vehicle width direction W of the pair of side edges **71c**.

[0113] A through hole **76**, which penetrates the coupling portion **73** in the front-rear direction L, is formed at each of four corner portions of the protruding portion **73a**.

<Support Member **80**>

[0114] As illustrated in FIGS. **4** and **5**, a second portion **87** of an extension portion **85** includes a plate-shaped base portion **88** and a guide portion **90** protruding rearward from the base portion **88**.

[0115] As illustrated in FIG. 5, the guide portion **90** includes a first wall portion **91**, a second wall portion **92**, an intermediate wall portion **95**, a third wall portion **96**, and a fourth wall portion **97**. [0116] The first wall portion **91** and the fourth wall portion **97** are formed at an interval in the vehicle width direction W, and protrude rearward from the base portion **88** along the front-rear direction L.

[0117] The second wall portion **92** extends inward in the vehicle width direction W and forward from a rear end of the first wall portion **91**. The second wall portion **92** is inclined such that the more inward a portion is located in the vehicle width direction W, the more forward the portion is located.

[0118] A rear surface of the second wall portion **92** is an inclined surface **92a** which is inclined with respect to both the front-rear direction L and the vehicle width direction W, and is in contact with the cutout portion **73b** from an inner peripheral side of the tubular portion **71**.

[0119] The third wall portion **96** extends outward in the vehicle width direction W and forward from a rear end of the fourth wall portion **97**. The third wall portion **96** is inclined such that the more outward a portion is located in the vehicle width direction W, the more forward the portion is located. A rear surface of the third wall portion **96** is an inclined surface **96a** which is inclined with respect to both the front-rear direction L and the vehicle width direction W, and is in contact with the cutout portion **73b** from an outer peripheral side of the tubular portion **71**.

[0120] That is, the guide portion **90** sandwiches the cutout portion **73b** from the inner peripheral side and the outer peripheral side of the tubular portion **71** by the inclined surface **92a** and the inclined surface **96a**.

[0121] The intermediate wall portion **95** couples a front end of the second wall portion **92** and a front end of the third wall portion **96** and extends in the vehicle width direction W. The intermediate wall portion **95** has a facing surface **95a** facing a front end of the cutout portion **73b**.
<Bumper Reinforcement **100**>

[0122] As illustrated in FIG. 4, the bumper reinforcement **100** includes a plate-shaped main body portion **101** extending in the vehicle width direction W, and an upper attachment portion **110** and a lower attachment portion **112** that are coupled to the main body portion **101**. The main body portion **101** includes a front wall portion **104** extending in the upper-lower direction Z, an upper wall portion **105** extending rearward from an upper end of the front wall portion **104**, and a lower wall portion **106** extending rearward from a lower end of the front wall portion **104**. An accommodating recessed portion **107** having a U-shaped cross section is formed on a rear surface **102** of the main body portion **101** by the front wall portion **104**, the upper wall portion **105**, and the lower wall portion **106**.

[0123] As illustrated in FIG. 5, the second portions **87** of the extension portions **85** are accommodated inside the accommodating recessed portion **107**. In the present embodiment, the front wall portion **104** and the second portion **87** are separated from each other in the front-rear direction L.

[0124] As illustrated in FIG. 4, the upper attachment portion **110** extends upward from a rear end of the upper wall portion **105**.

[0125] The lower attachment portion **112** extends downward from a rear end of the lower wall portion **106**.

[0126] On the upper attachment portion **110** and the lower attachment portion **112**, through holes **111** and through holes **113** are respectively formed at positions corresponding to the through holes **76** of the coupling portion **73**.

[0127] The bumper reinforcement **100** is attached to front ends of the pair of crash boxes **70** by screwing nuts **68** to respective bolts **67** inserted through the through holes **76** and **111** and bolts **67** inserted through the through holes **76** and **113**.

Functions and Effects of Second Embodiment

[0128] According to the present embodiment, the following functions and effects can be

implemented in addition to the functions and effects (1-1), (1-3), and (1-4) of the first embodiment. [0129] (2-1) The guide portion **90** has the pair of inclined surfaces **92a** and **96a** configured to sandwich the cutout portion **73b** from the inner peripheral side and the outer peripheral side of the tubular portion **71** in the vehicle width direction **W**, and the facing surface **95a** facing the front end of the cutout portion **73b**.

[0130] According to such a configuration, as illustrated in FIG. 6, the guide portion **90** is engaged with the cutout portion **73b** of the coupling portion **73**, so that the crash box **70** is guided to an attachment position along the inclined surface **92a** in the vehicle width direction **W**. Further, as illustrated in FIG. 5, the cutout portion **73b** is surrounded by the pair of inclined surfaces **92a** and **96a** and the facing surface **95a**, so that a relative movement of the crash box **70** with respect to the support member **80** is restricted. Therefore, the crash box **70** can be fixed at the attachment position.

[0131] (2-2) The rear surface **102** of the bumper reinforcement **100** has the accommodating recessed portion **107** configured to accommodate the extension portions **85**.

[0132] According to such a configuration, in the case where the extension portion **85** made of the resin is located between the coupling portion **73**, which is the front end of the crash box **70**, and the bumper reinforcement **100**, the extension portions **85** are accommodated in the accommodating recessed portion **107**. Accordingly, a compressive stress acting on the extension portion **85** sandwiched between the crash box **70** and the bumper reinforcement **100** can be reduced.

Therefore, creep deformation of the extension portion **85** is prevented. Therefore, the durability of the support member **80** can be improved.

<Modifications>

[0133] The present embodiment can be modified and implemented as follows. The present embodiment and the following modifications can be combined with each other and implemented without technical contradiction.

[0134] In the above-described first embodiment, the front surface of the base portion **38** of the extension portion **35** is located on the same plane as the flat portion **25** of the front end surface **24** of the coupling portion **23**, but the front surface of the base portion **38** may be located behind the flat portion **25**.

[0135] In the above-described first embodiment, the accommodating recessed portion **26** is formed on the front end surface **24** of the coupling portion **23**, and in the above-described second embodiment, the accommodating recessed portion **107** is formed on the rear surface **102** of the bumper reinforcement **100**, but the present invention is not limited to the this. The accommodating recessed portions **26**, **107** may be omitted as long as the creep deformation of the extension portions **35**, **85** caused by being respectively sandwiched between the front ends of the crash boxes **20**, **70** and the bumper reinforcements **50**, **100** is small.

[0136] The support members **30**, **80** may be made of the metal.

[0137] In the above-described embodiments, the support members **30**, **80** are each integrally molded, but a support body and an extension portion, which are separately formed, may be attached to each other.

[0138] In the above-described first embodiment, the two holes **27** into which the first guide portion **40** and the second guide portion **45** are respectively inserted are used as the engaged portions, but the engaged portions may be two apertures into which the first guide portion **40** and the second guide portion **45** are respectively inserted.

[0139] In the above-described first embodiment, the first guide portion **40** and the second guide portion **45** are inserted into the respective holes **27**, and in the above-described second embodiment, the cutout portion **73b** is sandwiched by the inclined surfaces **92a** and **96a** of the guide portion **90**, but the present invention is not limited to this. As illustrated in FIGS. 7 and 8, a pair of extension portions **235** of a support member **230** may respectively have a through hole **228** which extends in the front-rear direction **L** and through which a tubular portion **221** of a crash box

220 is inserted. As illustrated in FIG. 7, the through hole **228** has a rectangular cross section, and has a bottom surface **241** extending along both the front-rear direction L and the vehicle width direction W, a top surface **242** facing the bottom surface **241** in the upper-lower direction Z, and a pair of side surfaces **243** that are located on both sides in the vehicle width direction W and couple the bottom surface **241** and the top surface **242**. The bottom surface **241** and the top surface **242** are parallel to each other and sandwich the tubular portion **221** in the upper-lower direction Z. As illustrated in FIG. 8, the pair of side surfaces **243** are inclined such that a distance between the pair of side surfaces **243** in the vehicle width direction W increases as the pair of side surfaces **243** extend rearward (upper side in FIG. 8). The crash box **220** is guided to an attachment position indicated by a solid line in FIG. 8 by engaging a tip of the crash box **220** indicated by a two-dot chain line in FIG. 8 with inclined surfaces of the side surfaces **243**. Even in this case, the same effect as the above-described effect (1-1) can be implemented.

[0140] In the above-described embodiments, the inclined surfaces **42a**, **46a**, **47a**, **92a**, and **96a** are inclined with respect to both the front-rear direction L and the vehicle width direction W, but the present invention is not limited to this. For example, a guide portion may have an inclined surface which is inclined with respect to both the front-rear direction L and the upper-lower direction Z. In this case, a crash box is guided to an attachment position along the inclined surface of the guide portion in the upper-lower direction Z.

[0141] In the above-described embodiments, a pair of vehicle body frames to which the crash boxes **20** or the crash boxes **70** are attached are embodied as front side members, but the pair of vehicle body frames may be, for example, suspension members.

Claims

1. A front structure of a vehicle, comprising: a pair of vehicle body frames disposed at an interval in a vehicle width direction; a pair of crash boxes located in front of the vehicle body frames in a front-rear direction of the vehicle, and attached to the vehicle body frames respectively; a support member including a support body located between the pair of crash boxes, and a pair of extension portions that extend outward in the vehicle width direction from the support body and are located on both sides of the support body in the vehicle width direction; and a bumper reinforcement that extends in the vehicle width direction, and is attached to front ends of the pair of crash boxes, wherein the extension portion includes a guide portion configured to guide the crash box to an attachment position by engaging with a front end of the crash box, the guide portion having an inclined surface that is inclined with respect to the front-rear direction.
2. The front structure of a vehicle according to claim 1, wherein the crash box includes a tubular portion extending in the front-rear direction, and a coupling portion coupled to a front end of the tubular portion and protruding to an outer peripheral side of the tubular portion, a front end surface of the coupling portion has two engaged portions having a hole shape or an aperture shape and provided at an interval in the vehicle width direction, and two of the guide portions are inserted into the two engaged portions respectively.
3. The front structure of a vehicle according to claim 1, wherein the crash box includes a tubular portion extending in the front-rear direction, and a coupling portion coupled to a front end of the tubular portion, the coupling portion has a cutout portion not protruding outward in a radial direction of the tubular portion, and the guide portion has a pair of the inclined surfaces configured to sandwich the cutout portion from an inner peripheral side and an outer peripheral side of the tubular portion in the vehicle width direction, and a facing surface facing a front end of the cutout portion.
4. The front structure of a vehicle according to claim 1, wherein the support member including the support body and the pair of extension portions is integrally molded.
5. The front structure of a vehicle according to claim 4, wherein the support member is made of a

fiber reinforced resin.

6. The front structure of a vehicle according to claim 5, wherein the crash box includes a tubular portion extending in the front-rear direction, and a coupling portion coupled to a front end of the tubular portion and protruding to an outer peripheral side of the tubular portion, and a front end surface of the coupling portion has an accommodating recessed portion configured to accommodate the extension portion.

7. The front structure of a vehicle according to claim 5, wherein a rear surface of the bumper reinforcement has an accommodating recessed portion configured to accommodate the extension portion.
